

Energy System Model for Mauritius

This report outlines the methodology and data inputs used to build the energy system model for Mauritius. The model was designed to analyze the electricity sector from 2023 to 2050. The model utilizes the Open-Source Energy Modelling System (OSeMOSYS) tool to optimize capacity expansion with the least cost pathway. The primarily scenarios investigated include the following:

1. Business as usual (BAU) with the current policy intentions based on the government objectives, that is with introduction of liquified natural gas power plant. The growth of renewable energy follows a modest trend to reach 60% of production for renewable energy sources by 2035, in line with the national target.
2. Renewable energy transition using locally available renewable biomass sources such as cane tops and leaves, bagasse, eucalyptus, forest maintenance biomass, and refused derived fuels.
3. Coal phase out by 2030 with Accelerated renewable transition using the conventional renewable energy technologies, both locally available and imported biomass, solar PV, onshore wind, with battery energy storage systems.
4. Coal phase out by 2030 with introduction of liquified natural gas. This scenario models the potential of the global LNG fuel price being increased by 150% due to persistent high demand.
5. Coal phase out by 2030 with introduction of liquified natural gas. This scenario models the potential of the global LNG fuel price being increased by 150% due to persistent high demand, and global reduction in energy storage prices and renewable energy technology prices.

Objectives of the model

The primary objective of the energy model is to stimulate the long-term development of the electricity system of Mauritius. Results from the model will be used to determine the least cost capacity expansion plan to meet the future demand of the island, to assess different pathways to achieve the renewable energy targets and emission reduction commitments.

Model Horizon

The model was calibrated using the generation data to the year 2023. The modelling period spans from 2023 to 2050. The latter was chosen to ensure the robustness of the analysis. This time span avoids the distortions in energy demand caused by the pandemic-related shocks, while providing a long-term horizon to capture the complete coal phase-out by 2030 and to evaluate different pathways for the progression towards the 2035 renewable energy target of 60%. The model covers the close-loop Mauritius national grid system, and omits other islands in the Republic of Mauritius. The sectoral scope in this energy model focuses on the electricity sector only, which includes fuel supply, electricity generation, battery storage systems (catered for using the capacity factor) and, transmission and distribution combined as a single unit (since we do not have data on the efficiencies for each segment documented).

To capture the seasonal and daily variations in both electricity demand and renewable energy generation, the model was built to represent four distinct periods. The year was divided into the two seasons, summer and winter. Each season was then divided into two daily time brackets, day and night. This resulted into 4 time slices, Summer Day, Summer Night, Winter Day and Winter Night.

Table 1: Electricity Demand in GWh

Years	Electricity Demand (GWh)
2023	3266.11
2024	3418.06
2025	3728.06
2026	3796.39
2027	3864.44
2028	3932.50
2029	4000.28
2030	4068.06
2031	4135.83
2032	4203.33
2033	4270.56
2034	4337.78
2035	4405.00
2036	4472.22
2037	4539.17
2038	4606.39
2039	4673.06
2040	4740.00
2041	4806.67
2042	4873.61
2043	4940.28
2044	5006.67
2045	5073.33
2046	5139.72
2047	5206.39
2048	5272.78
2049	5339.17
2050	5405.28

Power plant inventory

The following tables illustrate the different characteristics and techno-economic parameters of existing power plants modelled. The techno-economic performance of the technologies is mainly inputted as the three primary cost parameters in OSeMOSYS. Capital cost represents the upfront investment needed per unit of capacity, in this case expressed in Million USD per Gigawatt (MUSD/GW), this parameter includes the conversion technology and the commissioning costs. The discount rate is used to annualize this expense during the course of the operational life of the technology. In this model, a discount rate of 5% was utilized. The fixed cost includes the expenses related to salaries, insurance, routine maintenance, and fixed cost is incurred whether or not the facility is producing electricity. While the variable cost consists of costs directly related to the amount of electricity generated, which includes the fuel cost, and other consumables. For renewable energy technologies such as solar, wind, hydro, the variable cost is zero as the fuel is available freely. While renewable technologies may have minimal variable costs for consumables, it was assumed to be negligible compared to the fossil fuel powered technologies.

The availability factor is another crucial input component. It represents the fraction of time the power plant is available to produce electricity, regardless of whether the power plant is needed or not. Availability factor accounts for the amount of time the power plant is down for planned maintenance and failures. Available factor is expressed as a percentage. In this model, the availability factor of the renewable technologies was assumed to be 1, as the planned outage is very negligible compared to the thermal power plants. For the thermal power plants, it was assumed that the maintenance would span for 30 days annually, which resulted an availability factor = 0.92.

The power plant efficiencies are important parameters as they refer to the ration of useful electricity output to the energy content of the fuel input. In this case, the efficiencies from literature were not considered for the exisiting technologies, as most of them have exceeded their lifespan. Therefore the efficiencies were calculated using the fuel use data from Statistics Mauritius and the generation information from the CEB annual reports.

Table 2: Powerplant characteristics (CEB, 2024)

Powerplant Name	ID	Fuel Type	Installed Capacity (MW)	Operational Life (Years)
Omnicanne St Aubin	PWRCOA001	Coal	33	40
Omnicanne La Baraque	PWRCOA002	Coal-Bagasse	90	40
Terragen	PWRCOA003	Coal-Bagasse	71	40
Alteo	PWRCOA004	Coal-Bagasse	37	40
Sotravac Landfill Gas	PWRLFG001	Landfill Gas	3	30
Solar PV CEB	PWRSOL001	Solar	6	30
Solar PV IPP	PWRSOL002	Solar	85	30
Solar PV SSDG	PWRSOL003	Solar	13	30
Solar PV MSDG	PWRSOL004	Solar	10	30
Solar PV Floating	PWRSOL005	Solar	2.06	30
Eole Wind Farm	PWRWND001	Wind	9.35	20
Hydro power plants	PWRHYD001	Hydro	61	100
CEB Nicolay Power Station	PWRKER001	Kerosene (Jet Fuel)	78	40
CEB St Louis Power Station	PWRHFO001	Heavy Fuel Oil	110	40
CEB Fort Victoria Power Station	PWRHFO002	Heavy Fuel Oil	109.6	40
CEB Fort George Power Station	PWRHFO003	Heavy Fuel Oil	140	40

Table 3: Costs for the fossil fuel powered technologies

Power Plant Type	Capital Cost (MUSD/GW)	Fixed Cost (MUSD/GW)	Variable Cost (MUSD/GW)
Coal Power Plant	2566	72	3.6
Coal/Bagasse Power Plant	1384	150	1.3
Kerosene Gas turbine	1800	35	1.6
HFO Thermal Power Plant	2162	35	4.2

Table 4: Capital costs for renewable technologies

Power Plant Type	Capital Cost in MUSD/GW)		
	2023	2040	2050
Landfill Gas Turbine	1200	1200	1200
Hydro Power Plant	2419	2419	2419
Solar PV Farm	1290	958	759
Utility			
Solar PV Residential	2574	1631	1004
Solar PV	1635	1163	891
Commercial			
Floating Solar PV	1613	1198	949

Table 5: Fixed cost for renewable energy technology

Power Plant Type	Fixed Cost in MUSD/GW)		
	2023	2040	2050
Landfill Gas Turbine	115	115	115
Hydro Power Plant	63	63	63
Solar PV Farm	22	19	16
Utility			
Solar PV Residential	28	19	13
Solar PV	18	14	12
Commercial			
Floating Solar PV	20	16	14

Table 6: Power plant efficiencies

ID	Efficiency (%)	Input Activity Ratio (MW)
PWRCOA001	28	3.576
PWRCOA002 (Mode 1: Coal)	16	6.394
PWRCOA002 (Mode 2: Bagasse)	20	4.954
PWRCOA003 (Mode 1: Coal)	24	4.103
PWRCOA003 (Mode 2: Bagasse)	18	5.603
PWRCOA004 (Mode 1: Coal)	20	5.097
PWRCOA004 (Mode 2: Bagasse)	10	9.585
PWRLFG001	33	3
PWRKER001	70	1.429
PWRHFO001	44	2.273
PWRHFO002	44	2.273
PWRHFO003	44	2.273

Future works

1. Use of a higher discount rate, potentially in the range of 8% to 15% to account for the events of risk of currency depreciation and high inflation. The addition of this sensitivity analysis would be beneficial in assessing how the renewable technologies deployment would be affected.
2. Refine the model to add hourly time slices, to understand how the demand load curves could be flattened, especially in summer season – which could have led to load shedding incidents in the past few months, when high temperatures increased cooling demand and renewable energy (solar) is not available.
3. Model battery energy storage systems and other types of storage to better support the ambitious renewable targets of Mauritius.
4. High renewable energy scenario with climate stress scenario, as a tropical island, Mauritius faces threats from cyclones, rising temperatures, extreme weather events and changing rainfall patterns due to climate change.

References

CEB (2024). *CEB Annual Report 2022-2023*. [online] Available at: https://ceb.mu/files/files/publications/Annual%20Report/CEB_Annual_Report_2022_2023.

Government of Mauritius (2024). *Digest of Energy and Water Statistics* [online] Available at: <https://govmu.org/EN/infoservices/energyutilities/Pages/energy.aspx>.